Normal Numbers: Theory and Proofs

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Abstract

This report brings together complementary perspectives normality. Beginning from definitions, base- r expansions, Pillai's characterization, Wall's uniform distribution criterion, and concrete examples of both normal and non-normal numbers, we establish Borel's Normal Number Theorem through distinct approaches: first, an elementary Rademacher and step-function proof that avoids Lebesgue integration altogether and second, Felix Hausdorff's constructive fourth-moment binomial argument in arbitrary integer bases $r \geq 2$.

1 Foundations, Equivalences, and Examples

1.1 Base-r Expansions and Definitions of Normality

Let $r \geq 2$ be an integer base. Every real number $\alpha \in \mathbb{R}$ admits a base-r expansion

$$\alpha = \sum_{j=-d}^{\infty} \alpha_{j,r} r^{-j}, \qquad \alpha_{j,r} \in \{0, 1, \ldots, r-1\}.$$

For a digit $b \in \{0, 1, \ldots, r-1\}$ and an integer $n \geq 1$, let $p_{b,n,\alpha,r}$ count how often b appears among the first n fractional digits of α:

$$p_{b,n,\alpha,r} = \big|\{j \in \mathbb{N}^+ : 1 \leq j \leq n \text{ and } \alpha_{j,r} = b\}\big|.$$

In the binary case $r = 2$ for $x \in [0, 1)$ written as $x = 0.d_1 d_2 d_3 \ldots_2 = \sum_{i=1}^{\infty} d_i 2^{-i}$, it is also handy to set $S_n(x) = p_{1,n,x,2} = \sum_{i=1}^{n} d_i$ for the number of ones among the first n digits.

Definition 1.1 (Simply Normal Number)**.** A real number α is *simply normal in base* r if every digit $b \in \{0, 1, \ldots, r-1\}$ occurs in its r-ary expansion with asymptotic frequency $\frac{1}{r}$:

$$\lim_{n \to \infty} \frac{p_{b,n,\alpha,r}}{n} = \frac{1}{r}.$$

In base 2, $x \in [0, 1)$ is simply normal when $\lim_{n \to \infty} \dfrac{S_n(x)}{n} = \dfrac{1}{2}$.

Simple normality only tracks single digits. Full normality demands much more: every finite word of digits must appear with the frequency expected from fair, independent rolls of an r-sided die.

Definition 1.2 (Normal and Absolutely Normal Numbers)**.** A real number α is *normal to base* r if for every integer $k \geq 1$ and every finite word $D_k = d_1 d_2 \cdots d_k \in \{0, 1, \ldots, r-1\}^k$ of length k, the block D_k occurs in the r-ary expansion of α with asymptotic frequency r^{-k}. Specifically, letting

$$A_r(D_k, n, \alpha) = \big|\{j : 1 \leq j \leq n - k + 1 \text{ and } \alpha_{j,r}\alpha_{j+1,r} \cdots \alpha_{j+k-1,r} = d_1 d_2 \cdots d_k\}\big|,$$

we require

$$\lim_{n\to\infty} \frac{A_r(D_k, n, \alpha)}{n} = \frac{1}{r^k}.$$

Moreover, α is *absolutely normal* if it is normal in every integer base $r > 1$.

Theorem 1.3 (Pillai, 1940)**.** *A real number α is normal in base r if and only if $r^m\alpha$ is simply normal in base r^n for every $n \in \mathbb{N}^+$ and every $m \in \mathbb{N}$.*

Remark. While Pillai's criterion via shifted powers $r^m\alpha$ in bases r^n and the overlapping block-frequency definition capture the exact same property, deducing one directly from the other requires delicate counting across residue offsets $m \pmod n$. In practice, the two viewpoints serve as independent, complementary tools.

Equivalent definition of normality- In his 1949 doctoral thesis, D. D. Wall recast normality in base $b \geq 2$ in via Weyl's uniform distribution modulo one.

Theorem 1.4 (Wall, 1949)**.** *Let $b \geq 2$ be an integer. A real number ξ is normal to base b if and only if the sequence $\left(\{b^n\xi\}\right)_{n\geq 1}$ is uniformly distributed modulo one; that is, for all $0 \leq u < v \leq 1$,*

$$\lim_{N\to\infty} \frac{\mathrm{Card}\{n : 1 \leq n \leq N,\ \{b^n\xi\} \in [u, v)\}}{N} = v - u.$$

To make precise the statement that "almost every" real number is normal, we recall the basics of null sets and Lebesgue measure on $[0, 1)$.

Definition 1.5 (Measure Zero and Almost All)**.** For a bounded interval J with endpoints $a < b$, let $\mu(J) = m(J) = b - a$ denote its length. A subset $A \subseteq [0, 1)$ has *measure zero* (or is a *null set*) if for every $\varepsilon > 0$ there exist intervals $J_1, J_2, \ldots$ covering A whose lengths sum to less than ε:

$$A \subseteq \bigcup_{n=1}^{\infty} J_n \qquad \text{and} \qquad \sum_{n=1}^{\infty} \mu(J_n) < \varepsilon.$$

We say that a property holds for *almost all* $x \in [0, 1)$ if the set of points where it fails has measure zero.

Lemma 1.6. *Every countable set is a null set, and any countable union of null sets is again a null set.*

Proof. If $A = \{x_1, x_2, \ldots\}$ is countable and $\varepsilon > 0$, enclose each x_n in an open interval J_n of length $\varepsilon/2^{n+1}$, so that $A \subseteq \bigcup_{n=1}^{\infty} J_n$ with $\sum_{n=1}^{\infty} \mu(J_n) = \varepsilon/2 < \varepsilon$. Similarly, if $A = \bigcup_{k=1}^{\infty} A_k$ is a countable union of null sets, covering each A_k by intervals $\{J_{k,n}\}_{n=1}^{\infty}$ of total length less than $\varepsilon/2^{k+1}$ gives a countable cover of A with total length strictly below ε. □

Theorem 1.7 (Borel, 1909)**.** *There exists a set $Z \subset \mathbb{R}$ of Lebesgue measure zero such that every $\alpha \in \mathbb{R} \setminus Z$ is absolutely normal.*

1.2 Examples of Non-Normal and Normal Numbers

Example 1.8 (Non-Normal Numbers)**.** (1) *Rational numbers:* Every $x \in \mathbb{Q}$ has an eventually periodic r-ary expansion (such as $\frac{1}{5} = 0.2000\ldots_{10}$ or $0.2437983983983983\ldots_{10}$). Once k exceeds the period length, only finitely many distinct blocks of length k can appear, so rational numbers are never normal.

(2) *Liouville constants:* Liouville's number $\sum_{n=1}^{\infty} 10^{-n!} = 0.11000100000000000000000001000\ldots_{10}$ and its variant $\sum_{n=1}^{\infty} n \cdot 10^{-n!} = 0.120003000000\ldots_{10}$ are transcendental, yet fail to be normal because increasingly long blocks of zeros overwhelm every other digit as $n \to \infty$.

Example 1.9. . For famous mathematical constants such as π and e, extensive numerical checks strongly point toward normality: across the first 20,000 decimal digits of π, all ten digits $0, 1, \ldots, 9$ appear evenly throughout the sequence, and among the first trillion hexadecimal digits of π, digit frequencies hover right around the expected count of $16^{-1} \times 10^{12} = 62{,}500{,}000{,}000$ (for instance, 62,499,613,666 occurrences). Even so, a rigorous proof that π or e is normal in any base remains completely open.

2 Proof I: Borel's Theorem Without Measure Theory

Our first proof shows that almost every $x \in [0, 1)$ is simply normal to base 2 using nothing beyond convergent numerical series and elementary integrals of step functions (functions that are constant on finitely many intervals), bypassing general Lebesgue integration completely.

2.1 Rademacher Functions and Fourth-Moment Orthogonality

Definition 2.1 (Rademacher Functions)**.** For each integer $n \geq 1$, partition $[0, 1)$ into 2^n dyadic subintervals $I_{n,k} = \left[\frac{k-1}{2^n}, \frac{k}{2^n}\right)$ for $k = 1, \ldots, 2^n$. The n-th *Rademacher function* $r_n : [0, 1) \to \{-1, +1\}$ alternates signs across these intervals:

$$r_n(x) = \begin{cases} +1 & \text{if } x \in \left[\frac{k-1}{2^n}, \frac{k}{2^n}\right) \text{ and } k \text{ is odd,} \\ -1 & \text{if } x \in \left[\frac{k-1}{2^n}, \frac{k}{2^n}\right) \text{ and } k \text{ is even.} \end{cases}$$

Proposition 2.2. *For every $x \in [0, 1)$, $r_n(x) = +1$ when the n-th binary digit d_n of x is 0, and $r_n(x) = -1$ when $d_n = 1$.*

Proof. When $x \in \left[\frac{k-1}{2^n}, \frac{k}{2^n}\right)$, the first n binary digits of x are the n-bit binary representation of the integer $k - 1$. Its last bit d_n is 0 if and only if $k - 1$ is even—meaning k is odd, which is precisely where $r_n(x) = +1$. Otherwise $d_n = 1$ and $r_n(x) = -1$. □

Set $R_n(x) = \sum_{i=1}^{n} r_i(x)$. Since each 0 among the first n binary digits adds $+1$ and each 1 adds -1, we have $R_n(x) = (n - S_n(x)) - S_n(x) = n - 2S_n(x)$, or

$$\frac{R_n(x)}{n} = 1 - 2\,\frac{S_n(x)}{n}.$$

Thus $S_n(x)/n \to \frac{1}{2}$ holds if and only if $R_n(x)/n \to 0$.

Lemma 2.3 (Orthogonality of Distinct Products)**.** *Whenever $1 \le i_1 < i_2 < \cdots < i_m$ are strictly increasing indices,*

$$\int_0^1 r_{i_1}(x)\, r_{i_2}(x) \cdots r_{i_m}(x)\, dx = 0.$$

Proof. Subdivide $[0,1)$ into the 2^{i_m-1} dyadic intervals $J = \left[\frac{k-1}{2^{i_m-1}}, \frac{k}{2^{i_m-1}}\right)$ of order $i_m - 1$. On each such J, the earlier functions $r_{i_1}, \ldots, r_{i_{m-1}}$ are constant, whereas r_{i_m} equals $+1$ on the left half of J and -1 on the right half. The integral over every J therefore cancels to 0, and summing over $k = 1, \ldots, 2^{i_m-1}$ gives 0. □

Because $r_i(x)^2 = 1$ everywhere on $[0,1)$, expanding $R_n(x)^4 = \sum_{i,j,k,l=1}^{n} r_i(x) r_j(x) r_k(x) r_l(x)$ yields n^4 terms that sort cleanly into five index patterns:

Case	Pattern (distinct indices)	Number of terms	Integral of each term
C-1	r_i^4	n	1
C-2	$r_i^2 r_j^2$	$3n(n-1)$	1
C-3	$r_i^3 r_j$	$4n(n-1)$	0
C-4	$r_i^2 r_j r_k$	$6n(n-1)(n-2)$	0
C-5	$r_i r_j r_k r_l$	$n(n-1)(n-2)(n-3)$	0

Summing the surviving terms gives

$$\int_0^1 R_n(x)^4\, dx = n + 3n(n-1) = 3n^2 - 2n,$$

and dividing by n^4 yields

$$\int_0^1 \left(\frac{R_n(x)}{n}\right)^4 dx = \frac{3n^2 - 2n}{n^4} = \frac{3}{n^2} - \frac{2}{n^3}. \tag{2.1}$$

This also explains why the fourth power is essential: the second moment only gives $\int_0^1 (R_n/n)^2\, dx = 1/n$, whose harmonic series diverges, whereas $\sum_{n=1}^{\infty}(3/n^2 - 2/n^3) < \infty$.

2.2 Beppo Levi's Monotone Convergence Theorem

When dealing with limiting operations under Lebesgue integration (such as summing infinite series of step functions or establishing tail bounds), the Monotone Convergence Theorem plays a foundational role.

Theorem 2.4 (Beppo Levi's Theorem / Monotone Convergence Theorem)**.** *Let $(f_n)_{n\ge1}$ be a sequence of non-negative measurable functions on a measurable set E such that $0 \le f_1(x) \le f_2(x) \le \cdots$ for almost all $x \in E$, and let $f(x) = \lim_{n\to\infty} f_n(x) = \sup_{n\ge1} f_n(x)$. Then*

$$\lim_{n\to\infty} \int_E f_n(x)\, dx = \int_E f(x)\, dx.$$

2.3 Step-Function Convergence Theorem

Lemma 2.5. *Let $(a_n)_{n\geq 1}$ be non-negative real numbers with $\sum_{n=1}^{\infty} a_n < \infty$. Then there exists a sequence of positive numbers $(b_n)_{n\geq 1}$ such that*

$$\lim_{n\to\infty} b_n = \infty \qquad \textit{and} \qquad \sum_{n=1}^{\infty} a_n b_n < \infty.$$

Theorem 2.6. *Let $(\varphi_n)_{n\geq 1}$ be real- or complex-valued step functions on $[0,1)$ satisfying*

$$\sum_{n=1}^{\infty} \int_0^1 |\varphi_n(x)|\, dx < \infty.$$

Then $\lim_{n\to\infty} \varphi_n(x) = 0$ for almost all $x \in [0,1)$.

Proof. Put $a_n = \int_0^1 |\varphi_n(x)|\, dx$. By Lemma 2.5, we can select $b_n > 0$ such that $b_n \to \infty$ and $\sum_{n=1}^{\infty} b_n \int_0^1 |\varphi_n(x)|\, dx < \infty$. Let $Z = \{x \in [0,1) : \varphi_n(x) \not\to 0\}$ be the set where convergence fails.

For any $x \in Z$, there exists $\eta > 0$ such that $|\varphi_n(x)| > \eta$ for infinitely many n. Because $b_n^{-1} \to 0$, we have $b_n^{-1} < \eta$ for all sufficiently large n, which means x lies in

$$A_n = \{y \in [0,1) : |\varphi_n(y)| > b_n^{-1}\}$$

for infinitely many n. Thus, for every integer $N \geq 1$, $Z \subseteq \bigcup_{k=N}^{\infty} A_k$.

Since φ_n is a step function, A_n is a finite union of disjoint intervals $J_1, \ldots, J_r$ of total length $\mu(A_n) = \sum_{i=1}^{r} \mu(J_i)$. On A_n we have $1 < b_n|\varphi_n(y)|$, so integrating over A_n gives

$$\mu(A_n) = \int_{A_n} 1\, dx \leq \int_{A_n} b_n|\varphi_n(x)|\, dx \leq b_n \int_0^1 |\varphi_n(x)|\, dx.$$

Hence $\sum_{n=1}^{\infty} \mu(A_n) < \infty$, so the tail lengths $\sum_{k=N}^{\infty} \mu(A_k)$ tend to 0 as $N \to \infty$. Given any $\varepsilon > 0$, choose N large enough that $\sum_{k=N}^{\infty} \mu(A_k) < \varepsilon$. The countable family of intervals forming $A_N, A_{N+1}, \ldots$ covers Z and has total length less than ε. Therefore Z has measure zero and $\varphi_n(x) \to 0$ for all $x \in [0,1) \setminus Z$. □

Proof of Simple Normality in Base 2. Define $\varphi_n(x) = \left(R_n(x)/n\right)^4 \geq 0$. Each φ_n is a step function, and (2.1) guarantees that $\sum_{n=1}^{\infty} \int_0^1 |\varphi_n(x)|\, dx = \sum_{n=1}^{\infty} (3n^{-2} - 2n^{-3}) < \infty$. Theorem 2.6 implies $\varphi_n(x) \to 0$ for almost all $x \in [0,1)$. Taking fourth roots gives $R_n(x)/n \to 0$, and hence $S_n(x)/n \to \frac{1}{2}$, for almost all $x \in [0,1)$. □

3 Proof II: Hausdorff's Constructive Binomial Proof

In 1914, Felix Hausdorff gave a direct, constructive proof of Borel's theorem for every integer base $r \geq 2$ using polynomial differential operators and a fourth-moment tail estimate.

3.1 Fourth-Moment Binomial Bound

Lemma 3.1 (Hausdorff, 1914)**.** *For every integer $r \geq 2$, there exists a positive constant $D > 0$ such that for all $n \in \mathbb{N}^+$,*

$$\sum_{p=0}^{n} \binom{n}{p} \frac{(r-1)^{n-p}}{r^n} \left(\frac{p}{n} - \frac{1}{r}\right)^4 \leq \frac{D}{n^2}.$$

Proof. Fix positive integers s and n, and define a sequence of two-variable polynomials recursively by

$$f_0(x,y) = \sum_{p=0}^{n} \binom{n}{p} x^{sp} y^{n-p}, \qquad f_{k+1}(x,y) = x\,\frac{\partial f_k}{\partial x}(x,y) - y\,\frac{\partial f_k}{\partial y}(x,y) \quad (k \in \mathbb{N}).$$

Applying the operator $x\partial_x - y\partial_y$ to each summand $x^{sp}y^{n-p}$ simply multiplies it by $sp-(n-p) = (s+1)p - n$. By induction on k, we obtain for every $k \geq 0$:

$$f_k(x,y) = \sum_{p=0}^{n} \binom{n}{p} \big((s+1)p - n\big)^k x^{sp} y^{n-p}. \tag{3.1}$$

On the other hand, the binomial theorem gives $f_0(x,y) = (x^s + y)^n$. Applying $x\partial_x - y\partial_y$ four times to $(x^s+y)^n$ produces a polynomial multiple of $(x^s+y)^{n-4}$, namely

$$f_4(x,y) = n(x^s+y)^{n-4} Q(x,y),$$

where grouping terms by powers of $(sx^s - y)$ yields

$$\begin{aligned} Q(x,y) = {} & 6n^2(s+1)^2 x^s y (sx^s - y)^2 + n^3 (sx^s - y)^4 \\ & + (s+1)^4 x^s y (x^{2s} - 4x^s y + y^2) + n(s+1)^3 x^s y (7(s+1)x^s y - 4s x^{2s} - 4y^2). \end{aligned}$$

Now set $r = s+1$, $x = r^{-1/s}$, and $y = (r-1)/r$. Then $x^s = 1/r$, so $x^s + y = 1$ and $sx^s - y = (r-1)/r - (r-1)/r = 0$. Both of the first two terms in $Q(x,y)$ vanish immediately, leaving the quadratic polynomial in n:

$$f_4(x,y) = 3(r-1)^2 n^2 + (r^3 - 7r^2 + 12r - 6)n.$$

Hence there exists a constant $C > 0$ independent of n with $f_4(x,y) \leq Cn^2$ for all $n \geq 1$. Substituting $x^s = 1/r$, $y = (r-1)/r$, and $s+1 = r$ into (3.1) at $k = 4$ gives

$$\sum_{p=0}^{n} \binom{n}{p} \frac{(r-1)^{n-p}}{r^n} (rp - n)^4 \leq Cn^2,$$

and dividing by $(rn)^4$ establishes the bound with $D = C/r^4$. □

3.2 Almost-Everywhere Absolute Normality

Fix a digit $b \in \{0, 1, \ldots, r-1\}$. Any fixed prefix of n digits in base r corresponds to a half-open r-adic interval of length r^{-n} (for example, in base 10, numbers starting with 314 form $[0.314, 0.315)$, which has length 10^{-3}). Among the r^n prefixes of length n, exactly $\binom{n}{p}(r-1)^{n-p}$

contain the digit b precisely p times, so the set of $\alpha \in [0,1)$ with $p_{b,n,\alpha,r} = p$ has Lebesgue measure $\binom{n}{p}(r-1)^{n-p}r^{-n}$.

For any $\varepsilon > 0$, consider the set of numbers whose digit frequency after n places deviates from $1/r$ by at least ε:

$$M_b(n,\varepsilon) = \left\{\alpha \in [0,1) : \left|\frac{p_{b,n,\alpha,r}}{n} - \frac{1}{r}\right| \geq \varepsilon\right\}.$$

Applying Markov's fourth-moment inequality together with Lemma 3.1 gives

$$\varepsilon^4\, m\big(M_b(n,\varepsilon)\big) = \sum_{\substack{p=0 \\ |p/n-1/r|\geq\varepsilon}}^{n} \binom{n}{p}\frac{(r-1)^{n-p}}{r^n}\varepsilon^4$$

$$\leq \sum_{p=0}^{n}\binom{n}{p}\frac{(r-1)^{n-p}}{r^n}\left(\frac{p}{n}-\frac{1}{r}\right)^4 \leq \frac{D}{n^2},$$

and therefore

$$m\big(M_b(n,\varepsilon)\big) \leq \frac{D}{\varepsilon^4 n^2}. \tag{3.2}$$

Theorem 3.2 (Borel's Theorem for Arbitrary Bases)**.** *For every integer $r \geq 2$, almost all real numbers are simply normal in base r, almost all real numbers are normal in base r, and almost all real numbers are absolutely normal.*

Proof. Let $M_b(\varepsilon) = \bigcap_{m=1}^{\infty} S_b(m,\varepsilon)$, where $S_b(m,\varepsilon) = \bigcup_{n=m}^{\infty} M_b(n,\varepsilon)$ consists of those $\alpha \in [0,1)$ that deviate by at least ε for some $n \geq m$. By (3.2),

$$m\big(S_b(m,\varepsilon)\big) \leq \sum_{n=m}^{\infty} m\big(M_b(n,\varepsilon)\big) \leq \frac{D}{\varepsilon^4}\sum_{n=m}^{\infty}\frac{1}{n^2} \xrightarrow{m\to\infty} 0.$$

Hence $M_b(\varepsilon)$ is a null set. The set M_b of numbers in $[0,1)$ where $p_{b,n,\alpha,r}/n$ fails to converge to $1/r$ is $M_b = \bigcup_{k=1}^{\infty} M_b(1/k)$, which is null by Lemma 1.6. Taking the finite union $\bigcup_{b=0}^{r-1} M_b$ shows that almost all $\alpha \in [0,1)$—and by translation invariance, almost all $\alpha \in \mathbb{R}$—are simply normal in base r.

To pass from simple normality to full normality in base r, invoke Pillai's theorem: α is normal in base r if and only if $r^m\alpha$ is simply normal in base r^n for all $n \geq 1$ and $m \geq 0$. Because $\mathbb{N} \times \mathbb{N}^+$ is countable and scaling by r^m preserves null sets, almost all real numbers are normal in base r. Finally, taking the countable intersection over all integer bases $r \geq 2$ and applying Lemma 1.6 shows that almost all real numbers are absolutely normal. □